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RESEARCH ARTICLE

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Effect of novel therapeutic medicine swertiamarin from *Enicostema axillare* in zebrafish infected with *Salmonella typhi*

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Abstract

Herbal treatments have been practiced by humans over centuries and therefore possess time-proven safety. However, it is crucial to evaluate the toxic effects of herbal medicine to confirm their safety, particularly when developing therapeutic drugs. Use of laboratory animals such as mice, rat, and rabbits was considered as gold standard in herbal toxicity assessments. However, in the last few decades, the ethical consideration of using higher vertebrates for toxicity testing has become more controversial. As a possible alternative model involving lower vertebrates such as zebra fish were introduced. Hence in the present study, swertiamain compound isolated from *E. axillare* was assessed for it antimicrobial activity in zebra fish larvae againt *S. typhi*. The cumulative mortality rate and bacterial localization in zebra fish larvae were studied. Biochemical markers assays were performed to find the preventive role of the compound during the typhoid infection. The results showed that zebra fish can be successfully used as a model to study typhoid infection and the anti-bacterial compound swertiamarin used in this study clears the bacterial load and pathogenic symptoms to a great extent.

KEYWORDS

anti-bacterial compound, biochemical markers assays, preventive assay, typhoid infection, zebra fish swertiamarin

1 | INTRODUCTION

Typhoid fever is being a silent health crisis in most developing and under-developed countries (Saha et al., 2002; Zavala Trujillo et al., 1991). Globally 21,7000,00 individuals are being affected by typhoid fever and 21,6510 deaths were reported (Crump et al., 2004; Hornick et al., 1970). In Asia, >100/100000 cases/year were reported accounting to 80% of world's population (Sharma et al., 2009) and high mortality was seen in school children (Crump et al., 2004). Typhoid is caused by *Salmonella typhi* a rod-shaped Gram-negative, non-spore forming and motile bacteria. Nearly 107 strains of *S. typhi* have been identified which shows difference in metabolism, virulence, and drug resistance that reasons behind the difficulty in treating typhoid (Deng et al., 2009; Parkhill et al., 2001). The emergence of multidrug resistance (MDR) further complexed the treatment and has become a supreme challenge in disease management (Sehra et al., 2013; Senthilkumar & Prabakaran, 2005). WHO currently recommends chloramphenicol, ampicillin and cotrimoxazole (trimethoprim-sulfamethoxazole), fluoroquinolones, third-generation cephalosporines (ceftriaxone, cefixime) and azithromycin for the treatment of enteric fever but unfortunately *S. typhi* strains were already resistant to almost all the three first-line recommended drugs for typhoid (Bhutta, 2006; Cleary, 2005; Kato et al., 2007; Kumar et al., 2008; Kumar & Gupta, 2007; WHO, 2003). Apart from Ty21a (oral) and Vi polysaccharide vaccines, Vi vaccine tetanus toxoid (Vi-TT) conjugate vaccine and Typbar-TCV vaccine have been recommended by WHO recently, but the efficacy needs to be checked (Milligan et al., 2018). Despite these drugs and vaccines, Antimicrobial Resistance (AMR) amongst *S. typhi* is getting worse (Browne et al., 2020). Swertiamarin belongs to group of compounds called secoirdoids and it plays key role in treating many human diseases (Leong et al., 2016). Swertiamarin was isolated and separated by methanol extraction and the purified compound was confirmed by spectral studies (Jaishree et al., 2009; Vishwakarma, 2004). Swertiamarin was reported in multiple diseases namely atherosclerosis, diabetes (Maroo et al., 2002; Maroo et al., 2003), antioxidant (Vaidya et al., 2009), anti-inflammatory (Kavimani & Manisenthlkumar, 2000), anti-nociceptive, hypertension (Bhatt et al., 2012), and rheumatoid arthritis (Saravanan, 2014). Because of its low toxicity and safe nature it has been used successfully as anti-spasmodic mediator in mice, pigeons, and rabbits (Suparna et al., 1998). The compound swertiamarin from *E. axillare* used in this study seems to have many medicinal benefits but it has not been still used for any bacterial diseases especially typhoid. This will be the first report to use swertiamarin to treat typhoid infection that too in a zebra fish model. Zebra fish (*Danio rerio*) infection models have been established for numerous bacterial, viral, and fungal pathogens (Gratacap & Wheeler, 2014; Masud et al., 2017; Varela et al., 2017) (Jaishree et al., 2009; Leong et al., 2016; Vishwakarma, 2004). Many infection models like *Salmonella typhimurium*, *Shigella flexneri*, *Pseudomonas aeruginosa*, *Burkholderia cenocepacia*, *Listeria monocytogenes*, *Staphylococcus aureus*, *Mycobacterium marinum*, *Mycobacterium abscessus*, and *Mycobacterium leprae* have been studied recently. Viruses like influenza A (Gabor et al., 2014), herpes simplex virus type 1 (Burgos et al., 2008), and chikungunya virus host-pathogen interactions were also thoroughly studied using zebra fish model (Palha et al., 2013). The reason for choosing zebra fish for our study is because of its well-established vertebrate model suitable for assessing developmental toxicity by exposing to toxicant during early development in the aquatic environment (Deng et al., 2009). From 36 h post-fertilization, primitive neutrophils appear in the embryo along with macrophages for first-line defense (Bennett et al., 2001; Willett et al., 2001). Also, approximately 84% of gene in the zebra fish seems to be associated with the human diseases (Howe et al., 2013; Truong et al., 2014). Previously, only fish-specific pathogens were studied in zebra fish but, in the present study, human pathogens were also assessed. Hence, in the present work, the mortality rate of the fish and the localization of the typhoid infection were studied in zebra fish model. Further, preventive assay was performed to find the anti-bacterial effect of the isolated swertiamain from *E. axillare* in typhoid.

2 | MATERIALS AND METHODS

2.1 | *S. typhi* infection

Glycerol stock of *S. typhi* from −80°C was plated freshly in LB medium and overnight culture was raised from the fresh plate. The overnight culture was serially diluted (1:10) to a count of 100 CFU per ml for the study.

2.2 | Test compound

Isolated and purified swertiamarin from *Enicostema axillare* was dissolved in double distilled water in the concentration of 10 mg/ml as stock (Perumal et al., 2022).

2.3 | Zebra fish housing and embryo collection

Wild type zebra fish (AB strain) were obtained from Live Tropics, Kadachanenthal, Madurai, Tamil Nadu, and maintained in clean well aerated glass tanks. Fishes were fed twice daily. Healthy adult male and female fishes (2:1) were maintained in light:dark cycle (14:10) to induce spawning. The embryos obtained were transferred into a clean Petri dish and washed with exposure medium to remove the debris. Then, the embryos were staged according to Kimmel et al. (Kimmel et al., 1995) and any unfertilized or dead embryos were removed. Zebra fish larvae were collected for the study.

2.4 | Exposure of zebra fish embryo and larvae to *S. typhi*

The exposure medium for the study contains 294 mg/L of $CaCl_2.2H_2O$, 63.0 mg/L of $NaHCO_3$, 123.3 mg/L of $MgSO_4{\cdot}7H_2O$ and 5.5 mg/L of KCl (ISO, 2007; Ku et al., 2015) which was prepared freshly before every exposure. 3dpf larvae were taken for the study. 100 CFU bacteria suspension was added to the titer plate containing larvae and injected into the embryo. Both larvae and embryo were observed at different time intervals. Triplicates were performed along with the control.

2.5 | Exposure of adult zebra fish to *S. typhi*

Three-month-old fish was taken for the study. Fish was starved for 24 h before starting the experiment. 100 CFU bacteria suspension was mixed with the fish feed and given to the fish. Control was maintained without infection.

2.6 | Acridine orange staining of infected embryo and larvae

Acridine orange (AO) staining was performed to find the localization of *S.typhi* infection induced in zebra fish larvae by following Mccarthy and Meseguer method (McCarthy & Senne, 1980; Meseguer et al., 1984). AO binds to nucleic acid and emits green fluorescence when bound to dsDNA. The larvae were exposed to AO (2.5 μg/ml)-stained *S. typhi* culture. After an hour, infected larvae were washed twice with PBS and anesthetized with 0.16% tricaine for examination under fluorescent microscope (Zhang et al., 2016).

2.7 | Preventive assay

Zebra fish larvae (3dpf) were placed in a 6-well titer plate containing 100 CFU of *S. typhi* culture. Swertiamarin stock of 10 mg/ml was prepared in double distilled water. After 3 h of infection, three different concentrations 15, 30, and 60 μg of swertiamarin prepared from the stock solution was added to the wells. Up to 12 h, larvae was observed. Both positive and negative controls were maintained.

2.8 | Liver sample preparation

All animal experiments were done at Madurai Kamaraj University, and it was reviewed by our in-house Institutional Animal Ethics Committee (IAEC Study No: 0211/Go190418/IAEC). Fish was anesthetized with 0.2% Tricaine and then euthanized by incubating in ice water for 15 min (Gupta & Mullins, 2010). The liver tissue was dissected from both swertiamarin-treated and control group fish. About 100 mg of liver tissue was homogenized and 1.0 ml of 0.1 M Tris- HCl buffer (pH 7.5) was added. The homogenized mixture was centrifuged at 10000 rpm at 4°C for 15 min, and the supernatant was utilized for biochemical enzymes and antioxidant activity analysis (Reitman & Frankel, 1957).

2.9 | Estimation of biomarker enzymes in adult fish liver

Glutamic oxaloacetic transaminases (AST) and glutamic pyruvic transaminases (ALT) enzymes were estimated by Reitman and Frankel method (Bennett et al., 2001). Lactate dehydrogenase (LDH) activity was calculated by Tietz method (Tietz, 1976). Na^+/K^+-ATPase activity was profiled using the protocol mentioned in Shiosaka (Shiosaka et al., 1971). Optical density (OD) was measured in UV-spectrophotometer.

2.10 | Statistical analysis

One-way ANOVA followed by Tukey's multiple comparison tests was performed in GraphPad Prism software (version 7) to determine the significance of the data.

3 | RESULT

3.1 | Exposure of zebra fish embryo and larvae to *S. typhi*

Zebra fish embryo exposed with 100 CFU of bacteria caused 100% mortality even before hatching (Figure 1). Within three hours, embryos got infection and dead (Figure 2). For zebra fish larvae, the mortality is directly proportional to the time of exposure. Up to 3 h, larvae count remained the same. From 3 h, the larvae count started to decline and by the end of 12 h all the larvae were dead (Figure 3). The morphological observations under the microscope showed peripheral edema, degenerated swim bladder, bent/curved spine, and degenerated body parts (Figure 4). The results showed that CNS of the larvae got seriously affected with the *S. typhi* infection causing major structural malformations.

3.2 | Exposure of adult zebra fish to *S. typhi*

Adult fish with *S. typhi* infection showed red spots on the body. Those spots might be because of the internal bleeding in the intestine and anal region (Figure 5). The fish also appeared swollen with upper skin discoloration which might be due to edema.

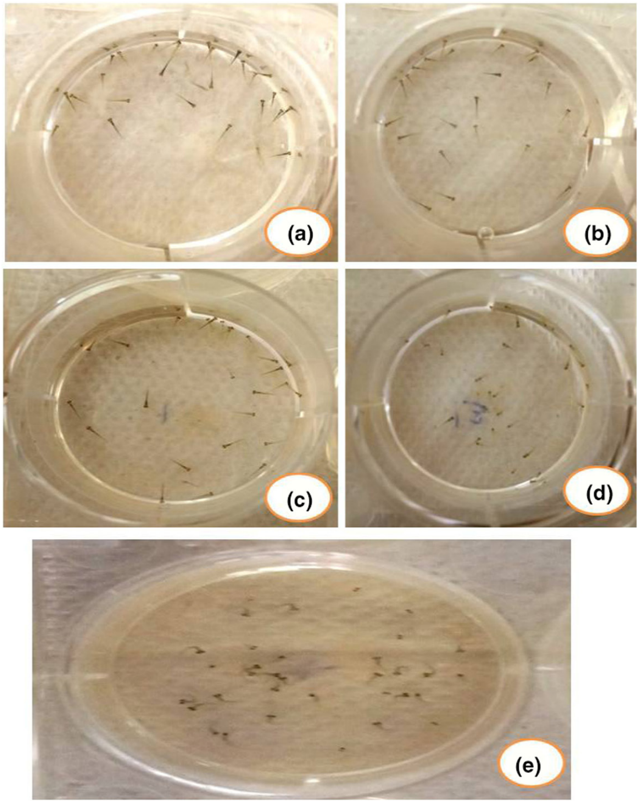

FIGURE 1 Mortality assay of zebra fish larvae exposed to *S. typhi* infection (a) Control (b) 3 h (c) 6 h (d) 9 h (e) 12 h

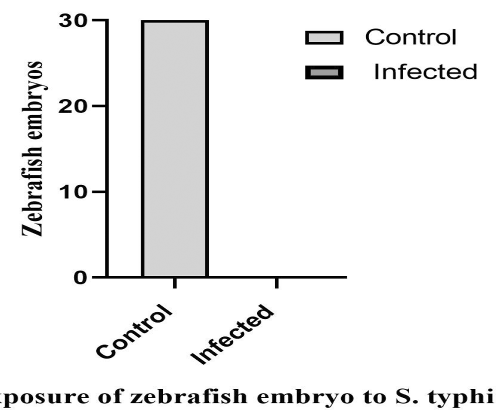

FIGURE 2 Mortality analysis of zebra fish embryo exposed to *S. typhi* infection

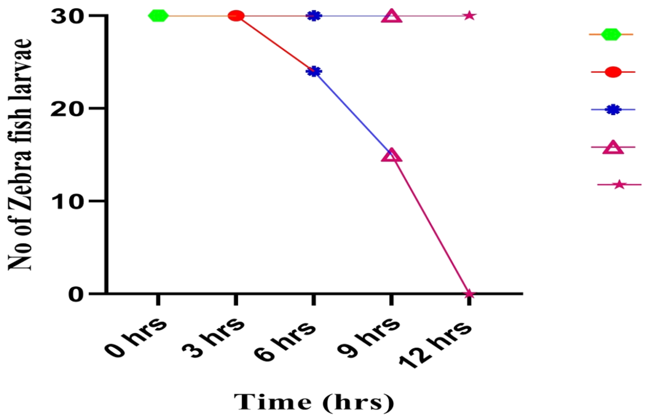

FIGURE 3 Survival rates of zebra fish larvae exposed to *S. typhi* infection

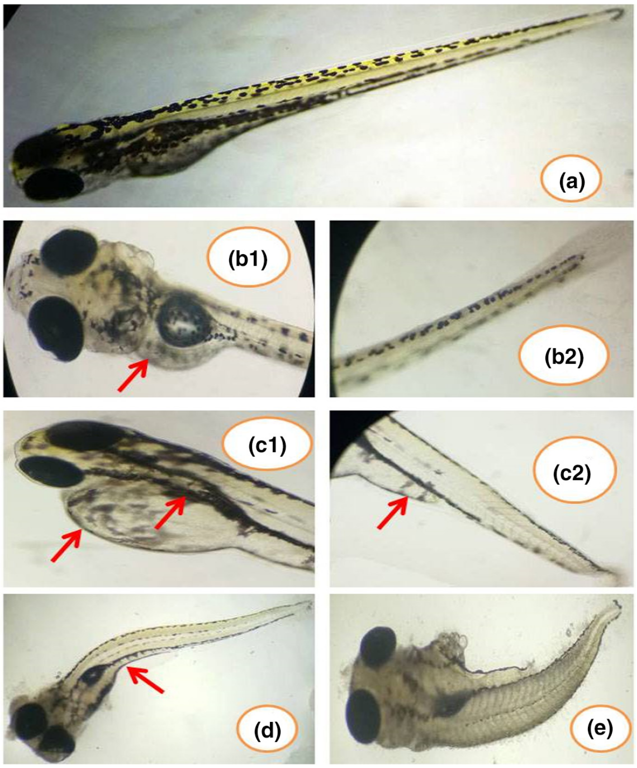

FIGURE 4 Morphological changes observed in zebra fish larvae exposed to *S. typhi* infection (a) control larvae (b1- peripheral edema and b2- degenerated tail part) (c1- swim bladder damage and c2- yolk sac edema) (d – changes in myotomes and degenerated swim bladder) (e – CNS damage and spinal deformation)

3.3 | Acridine orange staining of infected embryo and larvae

Bacterial load was located by staining with AO. Maximum load in the larvae was seen in the head region containing gills, heart, liver, brain, intestine, and kidney which extends till the full length of the tail (Figure 6). AO-stained embryo also showed similar pattern like larvae. When seen in fluorescent microscope, bacteria invaded the whole embryo causing death (Figure 7). The results showed that *S. typhi* shows infective symptoms affecting all the body parts covering nervous, circulatory, respiratory, and digestive system.

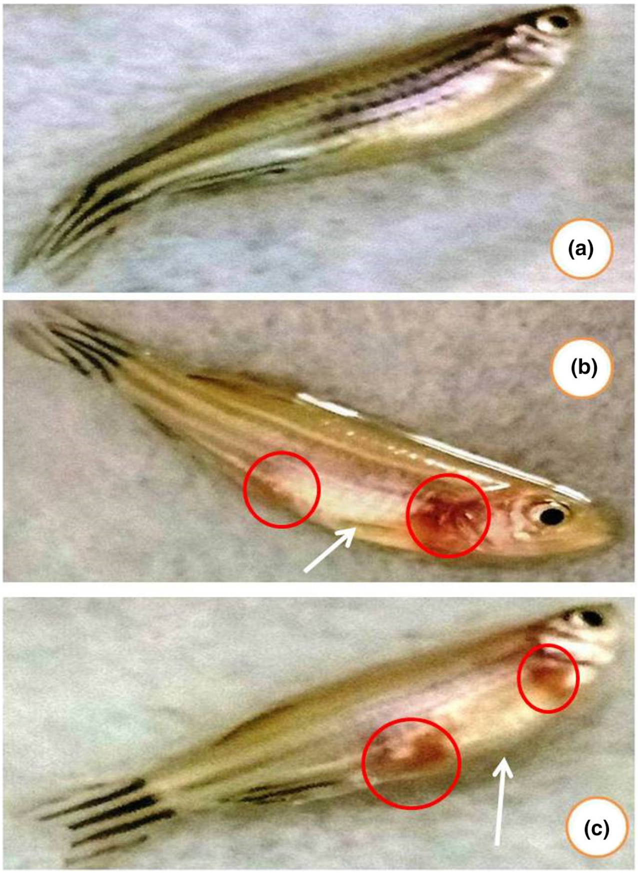

FIGURE 5 Exposure of zebra fish adult to *S. typhi* (a) control (b) *S. typhi*-infected female fish and (c) *S. typhi*-infected male fish

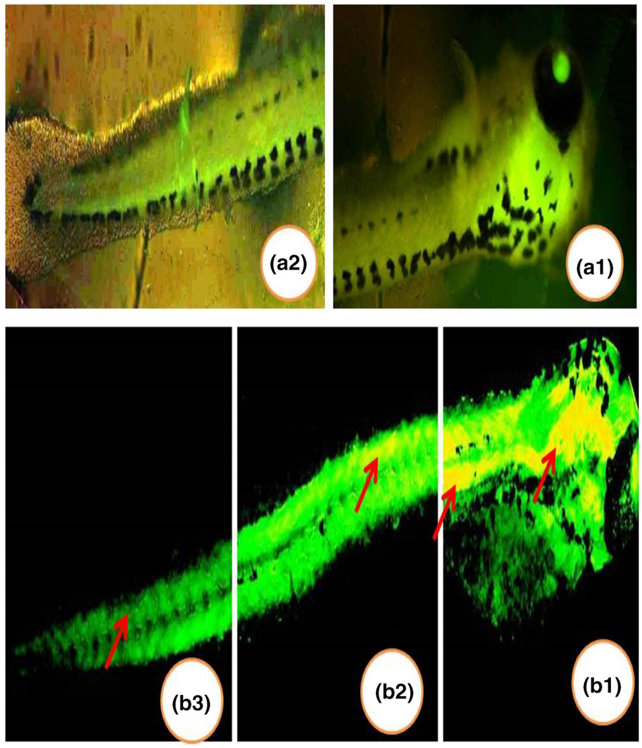

FIGURE 6 Acridine orange staining of the infected larvae showed the localization of the bacteria (a1) head (a2) tail region (b1) Head (b2) Abdomen (b3) Notochord regions

3.4 | Preventive assay

S. typhi infection starts to spread in the zebra fish larvae from 3 h and within 12 h the bacteria accumulates throughout the larvae body causing death. When the test compound swertiamarin added after 3 h of infection, the mortality rate seemed to reduce in 15 μg and 30 μg of concentration (Figure 8). At 60 μg concentration, only 10% of mortality was seen (Figure 9). The results clearly demonstrated that swertiamarin helps in clearing the accumulation of the bacteria inside the larvae, therefore preventing the appearance of symptoms and decrease in mortality. The 60 μg treated larvae were further grown to full fish for enzyme analysis.

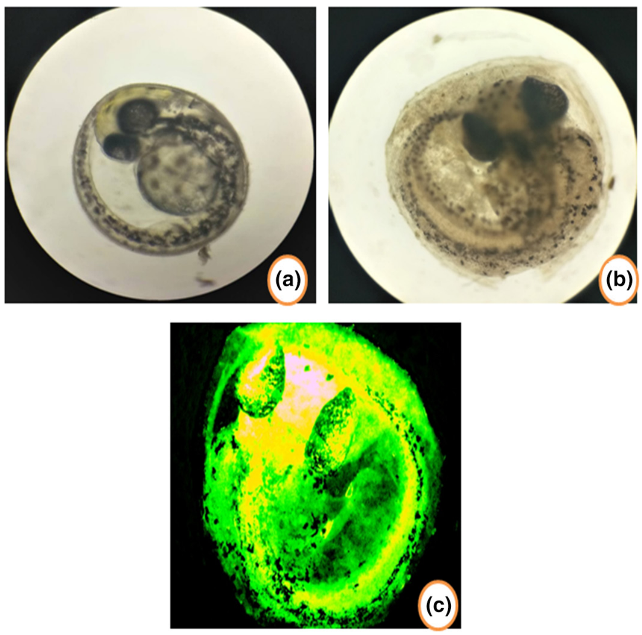

FIGURE 7 Acridine orange staining of infected embryo (a) Control (b) infected embryo (c) Acridine orange-stained infected embryo

3.5 | Estimation of biomarker enzymes in adult fish liver

Both infected and control fishes were taken for the study. The impact of Swertiamarin on AST, ALT, LDH, and Na^+/K^+-ATPase enzymes were shown in Figure 10. There was not much difference seen in AST, ALT, and LDH when compared to the control. The results indicate that only significant differences ($p < .05$) were observed. The decrease in liver Na^+/K^+-ATPase activity was found to be concentration-dependent.

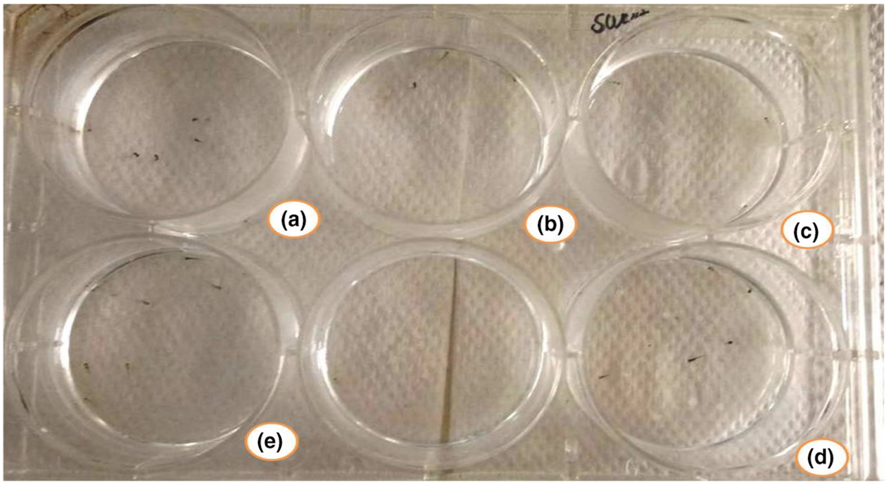

FIGURE 8 Preventive assay in zebra fish larvae (a) control (b) *S. typhi*-infected larvae (c) 15 μg swertiamarin-treated (d) 30 μg swertiamarin-treated (e) 60 μg swertiamarin-treated

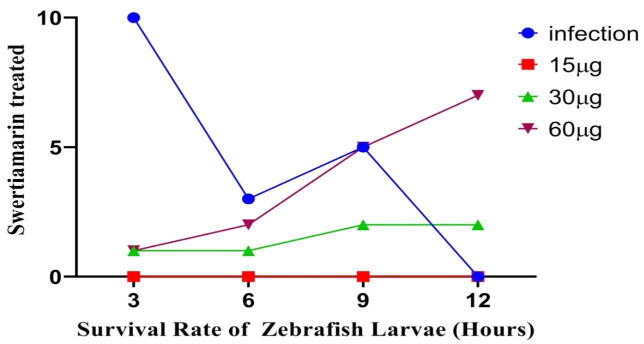

FIGURE 9 Survival rates of zebra fish larvae

4 | DISCUSSION

Antibiotic-resistant bacteria are increasing rapidly and because of its antimicrobial resistance (AMR) property 700,000 deaths are occurring per year (O'Neill, 2016). By 2050, AMR mortality may increase to 10,000,000 deaths annually (Kraker et al., 2016). In particular, gram-negative pathogens are a big alarm in multi drug resistance (MDR). From 2010, only five antibiotics agents against gram-negative pathogens has been approved by the FDA which is very less compared with the number of diseases caused by gram-negative bacteria (Koulenti et al., 2019). So, definitely there is a need to exploit more drugs to prevent humans from serious diseases.

Salmonella typhi is one among the gram-negative pathogens which takes a special place always because every year 11,000,000–20,000,000 people get typhoid globally in which 128,000 to 161,000 people die from it (https://www.coalitionagainsttyphoid.org/why-typhoid/drug-resistant-typhoid/; Acosta-Alonzo et al., 2020). By 2014 itself, 97 percent of typhoid patients resisted treatment from multiple drugs. In 2016, extremely drug resistant (XDR) typhoid fever outbreak occurred in the Sindh province of Pakistan (https://www.reactgroup.org/news-and-views/news-and-opinions/year-2019/outbreak-of-extremely-drug-resistant-typhoid-fever-in-pakistan/). While vaccines for *S. typhi* exist, they are only moderately effective (55–60%), limited protection duration, and are not suitable in young children. So, the population is almost at risk. The need and compulsion to find new drugs against typhoid are increasing day by day.

In our present study, natural compound swertiamarin from *E. axillare* was tested as an anti-bacterial agent against the *S. typhi* pathogen in zebra fish. For the study, zebra fish embryos were infected with *S. typhi* but it caused 100% mortality which shows that the pathogen is highly lethal to embryos. Acridine orange staining of the embryo shows the localization of the bacteria throughout the head and tail region showing the fatal effect. When zebra fish larvae were infected with *S. typhi* morphological changes were observed under the microscope. The mortality rate of the larvae seems to rise by increasing the time of exposure. The larvae after 3 h of infection edema started to appear. Within 12 h of infection, tail deformation, spinal curvature, pericardial edema, yolk sac edema, abdominal swelling, degenerated swim bladder, shrinked body, and changes in myotomes were seen. A study done in zebra fish larvae with influenza virus showed similar symptoms like arched back, yolk sac edema, and pericardial edema (Gabor et al., 2014). The deformity in notochord and myotomes in zebra fish larvae shows that *S. typhi* infection is associated with neurological symptoms involving central nervous system (CNS) and brain (Sejvar et al., 2012). The CNS infection might have caused harmful effects by negatively controlling the heart, kidney, liver, yolk sac, and swimming bladder. On the whole, typhoid causes abdominal, cardiovascular, neuropsychiatric, and respiratory symptoms (Paul & Bandyopadhyay, 2017). Acridine orange staining of the infected larvae compared with the control larvae shows the localization of the bacteria from head to abdomen region which extends through the notochord. The results show that the bacteria invade the CNS

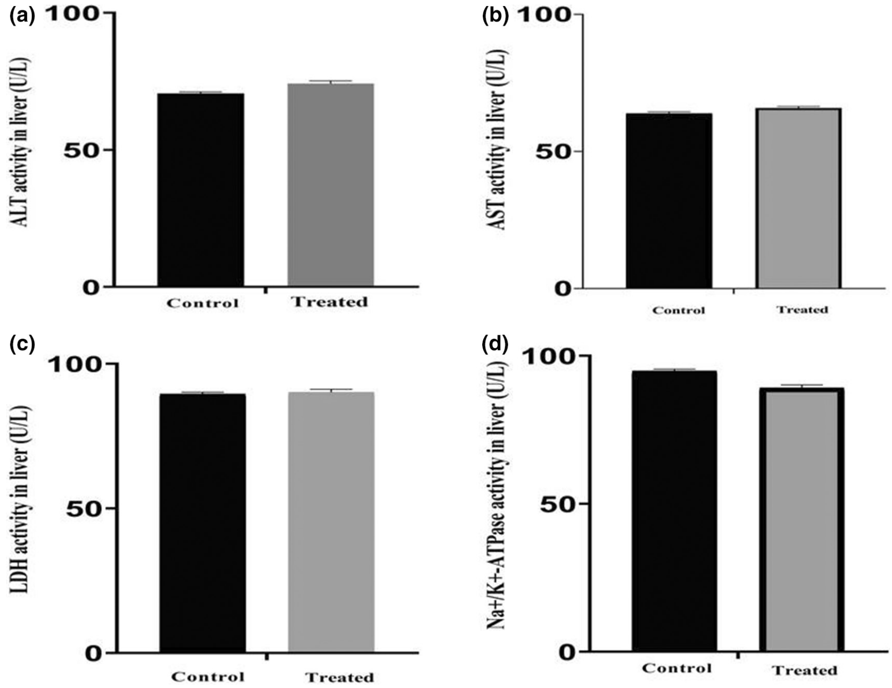

FIGURE 10 Estimation of biomarker enzymes in adult fish liver (a) ALT activity (b) AST activity (c) LDH activity (d) Na^{+}/K^{+}-ATPase enzymes

apart from giving common typhoid pathological symptoms. Typhoid infection undeniably has some impact on the developing larvae nervous system.

In adult zebra fish, *S. typhi* infection abdominal hemorrhage was noticed *(*Rodríguez et al., 2008*)*. Abdomen swelling and pale skin color was seen which might be due to the edema. Similar work done on *Vibrio parahaemolyticus* also showed abdominal hemorrhage in zebra fish (Zhang et al., 2017). Hemorrhage may be caused due to typhoid induced internal bleeding in the intestine (Ezzat et al., 2010).

The use of zebra fish larvae as a model for assessing the drug potency in typhoid research has not been fully exploited. Zebra fish has been used widely for drug screening because of the ease in studying CNS and cardiovascular system (Hill et al., 2012). In this study, preventive assay was performed with swertiamarin against *S. typhi* infection in three different concentrations. The results demonstrated that compare to 15 and 30 μg, 60 μg of swertiamarin showed 90% survival in zebra fish larvae. Spinal deformity and curvature were less in 60 μg treated larvae when compared to 15 and 30 μg. Other deformities like edema and abdominal swelling were not seen in the 60 μg treated ones. Further the enzymatic assays done on the fully grown 60 μg treated larvae showed that the AST, ALT, LDH, and Na^{+}/K^{+}-ATPase were statistically significant with the control. The overall results show that swertiamarin might help in preventing the zebra fish larvae against *S. typhi* infection to a great extent which indicates that the compound in future will definitely keep its foot in the field of medicine. Further exploitation of the compound as a drug has to be made.

5 | CONCLUSION

Zebra fish is a versatile animal model to study toxicology, drug discovery, and human diseases. In this study, zebra fish has been successfully proved to be an excellent model to study typhoid disease. Swertiamarin isolated from *E. axillare* showed anti-bacterial activity by reducing the symptoms in typhoid infected fish. The results from the study showed that swertiamarin seemed to reduce the *S. typhi* bacterial load in the zebra fish model. A low concentration of swertiamarin is efficient enough to act as drug. Further, analysis of the host-pathogen factors in zebra fish helps in prevention and treating of typhoid disease.

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This work was carried with own fund of first author.

CONFLICT OF INTEREST

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The datasets generated and/or analyzed during the current study are available from the corresponding author upon reasonable request.

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